

The Impact of Tick Size Reduction on Liquidity and Order Strategy: Evidence from the Jakarta Stock Exchange (JSX)

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Abstract

On January 3, 2005, the Jakarta Stock Exchange (JSX) implemented a new tick size of Rp10 in addition to the extant Rp5, Rp25, and Rp50 tick sizes. This new tick size affects shares within Rp500-Rp2000 price range, for these shares were previously traded with Rp25 tick size. The main purpose of adding this new tick size is to boost liquidity, especially for shares within the aforementioned range. Using daily data, t-tests, and cross-sectional multiple regressions, this study finds the new policy significantly reduces relative bid-ask spread, but also lessens bid and ask depth. From the aspects of width and immediacy cost, stock liquidity is enhanced; but from the viewpoint of bid and ask depth, stock liquidity is diminished. To resolve the two contradictory results, we use the ratio of average depth to relative spread (DRS). The new tick size does not trim down DRS significantly, or it does not diminish total liquidity. Furthermore, with smaller relative spread, our findings confirm traders' tendency to change their strategy from using limit orders to market orders, and to split their orders into smaller quantities.

We would like to thank Mohamed Ariff, Professor of Finance, Monash University, for his valuable inputs during the presentation of this paper in The 1st International Conference on Business and Management Research (ICBMR), in Bali, August 2006.

Keywords: Jakarta Stock Exchange, market microstructure, tick size, liquidity, bid-ask spread
JEL Classification: G14, G15, G18

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1. Introduction

The Jakarta Stock Exchange (JSX) is a computerized order driven market, operating a system known as JATS (*Jakarta Automated Trading System*) which was first implemented on May 22, 1995. On January 3, 2005, the JSX added a new tick size of Rp10 affecting shares within Rp500-Rp2000 price range, for these shares were previously traded using Rp25 tick. The JSX board of directors argued that, besides to enhance JSX revenue, the main purpose of adding this new tick size is to increase liquidity, especially for shares within the affected price range.

The effectiveness of tick size reduction on improving liquidity is still inconclusive. The proponents of the policy argue the reduction will benefit liquidity seeker because eventually the spread will decline. On the contrary, the challengers claim the decrease in spread will induce traders to use market orders instead of limit orders. Consequently the market depth, market transparency, and trade volume will diminish.

This paper investigates the impact of January 3, 2005 tick size reduction on: (1) stock liquidity, (2) traders order strategy, namely the tendency to use limit orders or market orders, and (3) traders average order size. Following Harris (2003), stock liquidity is defined as the stock ability to be traded quickly, cheaply, and without impacting its price. Liquidity is construed as having four dimensions: (1) width, (2) immediacy, (3) depth, and (4) resiliency.

Similar to Purwoto and Tandelilin (2004), liquidity is measured using end of day bid-ask spread and bid-ask depth. However, their studies were conducted around the July 3, 2000 implementation of a single Rp 5 market wide tick size.

2. Previous Studies

Harris (1994) is the first to investigate the impact of tick size reduction. He argues that tick size reduction will likely reduce bid-ask spread (lower transaction cost), and improve trade volume. On the other hand, if the tick size is too small, time priority rule will be ignored because of quote-matcher or front-runner problem. Quote-matchers will take advantage of large open orders position. They will try to place slightly better orders in front of the queued orders, which will be more profitable if the tick size is small. To protect themselves, informed traders will divide their orders to smaller quantities. Hence, they change from limit orders to market orders, and this change of strategy will decrease market depth.

Other studies, Ahn, Chao, and Choe (1996) found that tick size reduction in The American Stock Exchange in September 1992 diminished bid-ask spread, but did not increase transaction volume. Bacidore (1997) found that March 1997 tick size reduction in The Toronto Stock Exchange also reduced bid-ask spread and market depth, but trade activity did not change significantly. Bessembinder (2000) indicated that tick size reduction in NASDAQ lowered bid-ask spread and volatility. Goldstein and Kavajecz (2000) found declining bid-ask spread and market depth, following tick size reduction in the New York Stock Exchange in June 1997. Lau and McInish (1995) found tick size reduction in Singapore stock exchange in July 1994 reduced bid-ask spread. Bacidore et al. (2003) found that tick size reduction in The NYSE in 2000 caused traders

to reduce their order size, to hide some orders, and to cancel their limit orders more frequently. Bourghelle and Declerck (2004) found positive convex relationships between tick size and relative spread in the Paris Bourse.

Subsequent to July 3, 2000 tick size reduction in the JSX, Atmoko (2001), and Purwoto and Tandelilin (2004) conducted comparable research in this area. Atmoko (2001) found that tick size reduction is effective for reducing market spread for stocks below Rp500. Purwoto and Tandelilin (2004) found both market spread and market depth dropped significantly after the JSX implemented a single Rp5 tick size for all stocks.

3. Data and Methodology

This research utilizes the JSX stocks daily data from November 2, 2004 until February 28, 2005, with the event date on January 3, 2005. Since November 2, 2004 until December 30, 2004 (subperiod 1) stocks within the Rp500 to Rp2000 were traded with Rp25 tick, but from January 4, 2005 to February 28, 2005 (subperiod 2), these stocks were traded with the new Rp10 tick.

We only observe stocks listed on the main board and priced between Rp500 to Rp2000, for they are the ones affected by the new policy. To be included in the sample, stocks must meet the following criteria:

1. Closing price between Rp500-Rp2000 as of December 30, 2004.
2. The stock was traded at least once, during each subperiod.
3. The stock was not delisted nor split during the period.
4. The stock did not disburse any stock dividend nor issue rights during the period.

Using the said criteria, twenty four stocks were selected as the sample, representing eighty three percent of the total trade value of all stocks within Rp500 to Rp2000 price band (as of December 30, 2004).

Because the sample size is relatively small, we employ the non parametric sign test to study mean difference in two related samples (each subperiod is considered as one sample). In addition, we also use cross-sectional multiple regression to test the effect of tick size reduction on liquidity, while controlling other factors affecting liquidity as documented by previous research.

4. Results and Discussions

4.1. Bid-Ask Spread

One common measure of stock liquidity is its relative bid-ask spread (relative spread). Relative spread is used more often than nominal spread since it does not contain any currency unit, and it will be easier to compare between stocks and international markets with different currencies. The formula to calculate relative spread is presented in equation (1):

$$Relative\ Spread_{j,t} = \frac{A_{j,t} - B_{j,t}}{(A_{j,t} + B_{j,t}) / 2} \quad (1)$$

Relative Spread_{j,t} is relative bid-ask spread of stock j at time t; *A_{j,t}* is the best ask price of stock j at time t; and *B_{j,t}* is the best bid price of stock j at time t.

Table 1 shows the average relative spread of each stock in each subperiod. Before the new tick size, the average relative spread was 2.90 percent, while after the new policy, the average relative spread plummeted to 1.16 percent; or the average relative spread is 1.74 percent lower after the tick size reduction. To statistically test the mean difference before and after the tick size reduction, we use the t-test. The result of this test confirms that the mean of the relative spread is significantly lower after the tick size reduction (Table 2).

The reduction of relative spread means immediacy costs and transaction costs are lower after the tick size reduction. Based on immediacy and width dimensions of liquidity, lower spread means higher stock liquidity. If the relative spread is lower, than it is cheaper for investors who are in a hurry to transact (need immediacy) to use market order, or to buy at ask price or to sell at bid price. On the contrary, if the relative spread is higher, it will be more expensive for investors to transact immediately, or investors will tend to use limit order instead of market order.

The decline of relative spread is consistent with Porter and Weaver (1997) in Toronto Stock Exchange, Ronen and Weaver (1998) in American Stock Exchange, Ricker (1998) in New York Stock Exchange, Bessembinder (1997) in NASDAQ, Lau and McNish (1995) in Singapore Stock Exchange, and Purwoto and Tandelilin (2004) in the JSX.

Table 1. Average relative bid-ask spreads of all stocks in the sample, before and after January 3, 2005 tick size reduction, based on stocks ticker code.

No.	Stock Code	Relative Bid-Ask Spread		Change
		Before	After	
1	ADHI	3.24%	1.06%	-2.18%
2	AKRA	2.54%	1.66%	-0.88%
3	ANTM	1.49%	0.66%	-0.83%
4	AUTO	1.42%	1.13%	-0.29%
5	BBNI	1.66%	0.68%	-0.98%
6	BMRI	1.38%	0.61%	-0.77%
7	BUMI	3.29%	1.13%	-2.16%
8	CTRS	1.75%	0.86%	-0.89%
9	ENRG	4.36%	1.30%	-3.06%
10	GJTL	3.94%	1.38%	-2.56%
11	IATG	3.56%	1.26%	-2.30%
12	IDKM	3.70%	1.53%	-2.17%
13	INDF	3.39%	1.16%	-2.23%
14	INKP	2.70%	0.81%	-1.89%
15	KLBF	4.22%	1.45%	-2.77%
16	LPBN	3.89%	1.25%	-2.64%
17	LSIP	1.85%	0.72%	-1.13%
18	PGAS	1.72%	0.81%	-0.91%
19	PTBA	1.97%	0.64%	-1.33%
20	RALS	3.12%	1.34%	-1.78%
21	SMCB	3.16%	1.70%	-1.46%
22	SMRA	4.02%	1.45%	-2.57%
23	SUGI	2.94%	1.82%	-1.12%
24	TURI	4.40%	1.44%	-2.96%
Average		2.90%	1.16%	-1.74%

Table 2. Average relative bid-ask spreads, before and after January 3, 2005 tick size reduction.

Average Relative bid-ask spread		Average Change
Before ^a	After ^b	
2.90%	1.16%	-1.74%***

^aAverage relative bid-ask spread before tick size reduction (2 November 2004 to 30 December 2004)

^bAverage relative bid-ask spread after tick size reduction (4 January 2005 to 28 February 2005)

*** t-test result is significant at 1% level

4.2. Bid and Ask Depth

Another aspect of stock liquidity is its depth. A stock with higher depth is considered to be more liquid, because it could absorb higher transaction value before affecting its price. Like the bid-ask data, one of the limitations of this study is we only have end of day bid and ask depth (volume) data. In reality, bid-ask prices and volumes change continuously during trading hours.

Table 3. Averages of the best bid and ask depth, before and after January 3, 2005 tick size reduction.

	Average Volume (in million of shares)		Average Change (in million of shares)
	Before ^a	After ^b	
<i>Bid-Depth</i>	6.42	2.38	-4.04***
<i>Ask-Depth</i>	6.87	2.10	-4.77***

^aAverage volume (depth) before tick size reduction (2 November 2004 to 30 December 2004)

^bAverage volume (depth) after tick size reduction (4 January 2005 to 28 February 2005)

*** t-test result is significant at 1% level

After the new tick size is applied, Tabel 3 shows average bid-depth is 4.04 million shares or 62.92 percent lower, and 91.66 percent of stocks in the sample experience bid depth decline. Using non-parametric sign test this reduction is significant at one percent level. Similarly, average ask depth after the new policy is 4.77 million shares, or 69.43 percent lower. Using t-test, this reduction is also significant at one percent level.

The decrease of both relative bid-ask spread and bid-ask depth give differing signals on stock liquidity. Smaller relative spread means higher liquidity, but lower depth means lower liquidity. To settle these mix signals, we use the ratio of average depth to relative spreads.

4.3. Depth to Relative Spread

Harris (2003:394) explicates stock liquidity as the ability to trade large quantity quickly, cheaply, without impacting its price. Relative bid-ask spread measures quickness or immediacy, and the transaction cost. Lower relative spread means higher liquidity because it represents lower costs to transact immediately. Meanwhile bid and ask depth or the average of the two, measures the ability to absorb large trade quantity. More depth implies better liquidity, since it represents higher capability to absorb large trade without inducing large price change.

Since the sign tests confirm diminishing relative spread and depth, the impact of tick size reduction on liquidity is ambiguous. To measure trade-off between relative spread and depth, this study calculates depth-to-relative spread ratio (DRS) (Purwoto and Tandelilin, 2004) which is defined as:

$$Depth-to-Relative Spread (DRS)_{j,t} = \frac{(Ask Depth_{j,t} + Bid Depth_{j,t}) / 2}{Relative Spread_{j,t}} \quad (2)$$

$Ask Depth_{j,t}$ is the best ask volume for stock j at day t ; $Bid Depth_{j,t}$ is the best bid volume for stock j at day t ; and $Relative Spread_{j,t}$ is stock j relative bid-ask spread at day t . Intuitively this ratio measures whether the decrease of depth is larger or smaller than the decrease of relative spread. We use non-parametric sign test to check whether the change in DRS from before to after reduction period is significant, and the results are presented in Table 4.

After the new tick size, average DRS decreases 22,503,139 or 9.82 percent. Diminishing DRS means the decrease in depth is larger than the decrease in relative spread. Eventhough the average of DRS decreases, the t-test does not reject the null hypothesis, or the decrease is not statistically significant. This result suggests that the JSX tick size reduction does not impair overall stock liquidity.

This result is different from what Purwoto and Tandelilin (2004) revealed from the single Rp5 tick size policy of July 3, 2000. The policy affected all stocks in the JSX, and caused the DRS liquidity to plummet significantly. Meanwhile, the January 3, 2005 policy involves only certain stocks between Rp500 and Rp2000, and generates a more benign impact on stock liquidity.

Table 4. Average of Depth to Spread (DRS), before and after January 3, 2005 tick size reduction.

Average Depth-to-Spread (DRS)		Average Change ^c
Before ^a	After ^b	
229,217,733	206,714,594	-22,503,139 ^c

^aAverage depth-to-spread before tick size reduction (2 November 2004 to 30 December 2004)

^bAverage depth-to-spread after tick size reduction (4 January 2005 to 28 February 2005)

^c t-test result is not statistically significant

4.4. Multiple Regression Models

The nonparametric tests show the tick size reduction leads to statistically significant lower relative bid-ask spread and bid-ask depth, but does not significantly reduce the ratio of average depth to relative spread. Based on previous studies such as Porter and Weaver (1997), Ronen and Weaver (1998), Purwoto and Tandelilin (2004), and also Ekaputra (2006), we discover stock price, trade activity, and return volatility influence liquidity. Trade activity surrogates utilitarian traders' interests, or traders who are not primarily driven by profit; and return volatility represents the degree of market information asymmetry. To control these factors, and to study the impact of tick size reduction on both relative spread and depth to relative spread, we use the following multiple regression models:

$$\begin{aligned} \ln RS_i = & \beta_0 + \beta_1 (\ln PRICE_i) + \beta_2 (\ln FREQ_i) + \beta_3 (\ln VOLATILITY_i) + \\ & \beta_4 D * \ln PRICE_i + \beta_5 D * \ln FREQ_i + \beta_6 D * \ln VOLATILITY_i + \beta_7 D + \varepsilon_i \end{aligned} \quad (3)$$

$$\begin{aligned} \ln DRS_i = & \beta_0 + \beta_1 (\ln PRICE_i) + \beta_2 (\ln FREQ_i) + \beta_3 (\ln VOLATILITY_i) + \\ & \beta_4 D * \ln PRICE_i + \beta_5 D * \ln FREQ_i + \beta_6 D * \ln VOLATILITY_i + \beta_7 D + \varepsilon_i \end{aligned} \quad (4)$$

In model (3) RS is the relative bid-ask spread, while in model (4) DRS is the average depth to relative bid-ask spread. For both model (3) and (4), PRICE is the average stock price, FREQ is the average trade frequency, VOLATILITY is the stock return volatility, and D is a dummy variable which is equal to zero for pre tick size reduction (subperiod 1), and is equal to 1 for post tick size reduction (subperiod 2).

The hypothesis for model (3):

H1: Price will have a negative impact on relative spread. The higher the price the lower the relative spread

H2: Frequency will have a negative impact on relative spread. The higher the trade frequency the lower the relative spread

H3: Return volatility will have a positive impact on relative spread. Higher return volatility will increase the relative spread.

H4: Tick size reduction will affect the impact of price on relative spread

H5: Tick size reduction will affect the impact of trade frequency on relative spread

H6: Tick size reduction will affect the impact of return volatility on relative spread

H7: Tick size reduction will reduce the unconditional mean of relative spread

The hypothesis for model (4):

H1: Price will have an impact on depth to relative spread.

H2: Frequency will have a positive impact on depth to relative spread. The higher the trade frequency the higher the depth to relative spread

H3: Return volatility will have a negative impact on depth to relative spread. Higher return volatility will decrease the depth to relative spread.

H4: Tick size reduction will affect the impact of price on depth to relative spread

H5: Tick size reduction will affect the impact of trade frequency on depth to relative spread

H6: Tick size reduction will affect the impact of return volatility on depth to relative spread

H7: Tick size reduction will increase the unconditional mean of depth to relative spread

For both model (3) and (4), if the new tick size policy improves stock liquidity, we expect the data to support hypothesis number 7 (H7). The summary of expected signs of all regression coefficients are provided in Table 5, while the regression results are shown in Table 6.

Table 5. Expected signs of each coefficient in Model 3 and Model 4.

Variable	Coefficient	Model 3	Model 4
		Expected Sign	Expected Sign
Intercept	β_0	+	+
Ln_PRICE	β_1	-	+/-
Ln_VOLATILITY	β_2	-	+
Ln_FREQ	β_3	+	-
D*LN_PRICE	β_4	+/-	+/-
D*LN_VOLATILITY	β_5	+/-	+/-
D*LN_FREQ	β_6	+/-	+/-
D	β_7	-	+

Table 6. Regression results of Relative Spread and Depth to Relative Spread on price, volatility, frequency, and dummy variables.

	Dependent Variable			Dependent Variable		
	Ln_Relative Spread (Ln_RS)			Ln_Depth to Relative Spread (Ln_DRS)		
Variable	Coefficient	t-Statistic	Prob. (2-tail)	Coefficient	t-Statistic	Prob. (2-tail)
Intercept	3.04047	5.56965	0.00000	19.88575	5.95323	0.00000
Ln_PRICE	-0.75915	-12.23052	0.00000	-2.20920	-6.07141	0.00000
Ln_VOLATILITY	-0.05631	-1.46626	0.15080	1.18867	6.19989	0.00000
Ln_FREQ	0.32957	2.28745	0.02780	-0.51760	-0.76208	0.45070
D*LN_PRICE	0.14537	1.72959	0.09180	0.47128	0.91894	0.36390
D*LN_VOLATILITY	-0.04813	-0.90704	0.37010	0.21441	0.80998	0.42300
D*LN_FREQ	-0.28215	-1.73322	0.09120	0.07369	0.09717	0.92310
D	-2.54409	-3.59153	0.00090	-4.78044	-1.09444	0.28070
	Adjusted R-squared		0.93975	Adjusted R-squared		0.76443

Based on Table 6, especially for the coefficient of dummy variable (D), we find the tick size reduction significantly decreases relative bid-ask spread (RS), but does not affect average depth to relative spread (DRS). The results are consistent with the non-parametric tests conducted earlier. Hence, from the viewpoint of width and cost of immediacy, we can conclude that the new policy improves stock liquidity.

On the other hand, if we look at the more complete measure of liquidity represented by average depth to relative spread (DRS), the new tick size does not significantly affect the total stock liquidity. This finding is different from Purwoto and Tandelilin (2004), where they found July 3, 2000 tick size reduction actually diminished overall stock liquidity. The contrary results are probably caused by the scope of the two policies. The July 3, 2000 course of action affected all stocks regardless of their prices, while January 3, 2005 policy affects only stocks between Rp500 to Rp2000 price band.

Based on the findings, we could conclude that the January 3, 2005 decision made by the JSX board of directors is more selectively implemented, thus it achieves better results in improving stock liquidity.

4.5. Order Submission Strategy

A tick size reduction is likely to decrease relative bid-ask spread. However, if the relative spread is too narrow, the time priority rule becomes less meaningful, and the quote-matcher problem will rise. Quote-matching is a front-running strategy in which quote-machers try to trade in front of large limit order traders or ‘stepping ahead of the book’ (Ronen and Weaver, 1998). The relative spread therefore is the price that quote-machers must pay to front-run large traders submitting limit orders.

A narrow percentage spread makes quote-matching strategy more profitable. Therefore, informed traders will defend themselves from quote-machers by splitting their orders and by switching their trade strategies from limit-order to market-order. Subsequently, these responses will lead to lower market depth.

To analyze the inclination to alternate order submission strategy from limit order to market order, or in the JSX case, to marketable limit order¹, this study calculate the ratio of marketable limit orders to total orders (MALTO) defined as:

$$\text{Marketable Limit Orders to Total Orders (MALTO)}_{j,t} = \frac{\text{Marketable Limit Order}_{j,t}}{\text{Total Order}_{j,t}} \quad (5)$$

where *Marketable Limit Order*_{*j,t*} is the volume (number of shares) of marketable limit order for stock *j* at day *t*, and *Total Order*_{*j,t*} is the volume (number of shares) of total order for stock *j* at day *t*. Higher *MALTO* means more orders are submitted using marketable limit order. The non-parametric sign test is used to test whether *MALTO* from pre to post reduction period are significantly different, and the results are given in Table 7.

The average MALTO of the 24 stocks in the sample increases 4.90 percent from pre to post reduction period. The t-test strongly rejects the null hypothesis that there is no change in the MALTO average. These results suggest that traders tend to switch from limit orders to market orders after the tick size reduction.

¹ A buy (sell) marketable limit order has a limit price that is greater (less) than the best offer (bid) price. Because marketable limit orders are immediately executable, they resemble market orders.

Table 7. Average marketable limit orders to total orders (MALTO) before and after January 3, 2005 tick size reduction.

Average MALTO		Average Change
Before ^a	After ^b	
12.49%	17.39%	4.90%***

^aAverage MALTO before tick size reduction (2 November 2004 to 30 December 2004)

^bAverage MALTO after tick size reduction (4 January 2005 to 28 February 2005)

*** t-test result is significant at 1% level

4.6. Order Size

To prevent losses to quote-matchers, limit order traders must control how they expose their orders. Large traders often defend themselves from front-runners or quote-matchers by not showing the full extent of their interests through their limit orders. They tend to split their orders into smaller parts, or they display only a fraction of total orders. To investigate, this study calculates the ratio of total buy (sell) order quantity to total number of orders:

$$Buy (Sell) Order Size_{j,t} = \frac{\sum_{i=1}^{NO_{j,t}} Buy (Sell) Order Quantity_{j,t}}{NO_{j,t}} \quad (6)$$

Buy (Sell) Order Size_{j,t} is the of total buy (sell) order quantity to total number of orders; *Buy (Sell) Order Quantity_{j,t}* is the total volume (number of shares) of buy (sell) orders for stock *j* at day *t*; and *NO_{j,t}* is the number of orders for stock *j* at day *t*. The non-parametric sign test is then used to test whether the change in average order size from before to after reduction period is significant, and the results are shown in Table 8.

The average Buy Order Size decreases 66,615 shares or 33.25 percent from before to after reduction period. The average Sell Order Size decreases 35,467 shares or 21.47 percent from pre to post reduction period. The t-test strongly rejects the null hypothesis that there is no change in average Order Size. These results suggest that traders tend to split their orders after the tick size reduction.

Table 8. Average buy order size and sell order size, before and after tick size reduction.

	Average Volume per order (in number of shares)		Average Change (in number of shares)
	Before ^a	After ^b	
<i>Buy Order Size</i>	200,308	133,693	-66,615***
<i>Sell Order Size</i>	165,156	129,689	-35,467***

^aAverage buying order size before tick size reduction (2 November 2004 to 30 December 2004)

^bAverage selling order size after tick size reduction (4 January 2005 to 28 February 2005)

*** t-test result is significant at 1% level

5. Concluding Remarks

This study was triggered by the Jakarta Stock Exchange (JSX) decision on January 3, 2005 to implement a new tick size of Rp10, in addition to the extant three tick sizes of Rp5, Rp25, and Rp50. This new tick affects shares within Rp500-Rp2000 price range, for they were previously traded with Rp25 tick. The JSX board of directors claims that besides to enhance JSX revenue, the main purpose of adding this new tick size is to increase liquidity, particularly for shares within the aforementioned range.

This study measures liquidity using relative bid-ask spread, bid-ask depth, and the ratio of average depth to relative bid-ask spread. From the width and immediacy aspects, the implementation of Rp10 tick size improves stock liquidity, since relative bid-ask spreads or the costs of immediacy are significantly lower. On the contrary, from the depth viewpoint, stock liquidity is diminished, because bid and ask depth tumbled significantly. These two results are consistent with the findings of Purwoto and Tandelilin (2004), investigating the impact of the JSX tick size reduction on July 3, 2000.

To resolve the two contradictory results, we use the ratio of average depth to relative spread (DRS). We find the tick size reduction does not instigate significant change of average DRS, or it does not impair total liquidity. This result is different from Purwoto and Tandelilin (2004) since they found significant reduction in DRS after the adoption of a single tick size policy. We deduce the difference is rooted in the scope of the two events. On January 3, 2005, the new Rp10 tick size is added amid the existing three tick sizes of Rp5, Rp25, Rp50; and this course of action only affects certain stocks between Rp500 to Rp2000 price array. While, on July 3, 2000, the Rp5 single tick size was applied to all listed stocks regardless of their prices.

Based on the results, we could acknowledge that the JSX board of directors January 3, 2005 policy improves liquidity, in particular for shares affected by the new tick size. The board manages to reduce relative bid-ask spreads without demolishing the total liquidity, measured by the ratio of average depth to relative bid-ask spread (DRS).

Plummeting relative spreads mean more risks to large limit order traders and informed traders. Our findings show that the tick size reduction causes traders to change their order strategy. With

narrower relative spreads, informed traders will likely change their strategy from using limit orders to market orders or marketable-limit orders. This change of strategy is to protect them from parasitic traders, namely quote matchers and front runners.

For the same reason, lower relative spreads also significantly reduce limit order size, because informed traders must control how they expose their orders. Large traders often do not show the full extent of their interests through their limit orders. They are compelled to divide their orders into smaller parts, or they only display some portions of their total orders. Consequently, this strategy will trigger lower bid and ask depth.

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